Deploy application

Slide 1 — Deploy Application

Application Deployment

- Deployment makes the application available to users over the internet.
- The deployment platform depends on the application type and requirements.
- Common options:
    - **Frontend:** Streamlit Community Cloud
    - **Backend:** Render, Railway, Fly.io, or VPS
    - **Other options:** Deta and similar cloud platforms

Slide 2 — Render

Deploying with Render

- Render is a cloud platform for deploying web applications and APIs.
- Supports popular programming languages and Git-based deployments.
- Basic deployment steps:
    1. Connect GitHub repository.
    2. Select the project.
    3. Configure build and start commands.
    4. Add environment variables.
    5. Deploy the application.
- Provides automatic deployment when code changes are pushed.

Slide 3 — Railway

Deploying with Railway

- Railway provides a simple platform for deploying applications and backend services.
- Can deploy directly from a GitHub repository.
- Supports environment variables and databases.
- Basic process:
    - Connect repository
    - Configure project
    - Add required variables
    - Deploy
    - Monitor application logs

Slide 4 — Streamlit Community Cloud

Frontend Deployment

- Streamlit Community Cloud is suitable for deploying Streamlit-based Python applications.
- Connect the GitHub repository to the Streamlit platform.
- Select the required Python file.
- Configure dependencies using `requirements.txt`.
- Deploy and obtain a public application URL.

Example:

```
app.py
requirements.txt
```

Slide 5 — Fly.io / VPS for Backend

Backend Deployment Options

Fly.io

- Suitable for deploying backend applications in containers.
- Supports scalable and globally distributed deployments.
- Useful when Docker-based deployment is required.

VPS

- A Virtual Private Server provides greater control over the deployment environment.
- Install the required runtime, dependencies, web server, and database.
- Suitable for applications requiring customized server configuration.

Slide 6 — Deployment Workflow

Complete Deployment Process

```
Develop Application
        ↓
Prepare Dependencies
        ↓
Push Code to GitHub
        ↓
Choose Deployment Platform
        ↓
Configure Environment Variables
        ↓
Deploy Application
        ↓
Test & Monitor
```

Platform Selection:

- **Streamlit Cloud** → Streamlit frontend
- **Render / Railway** → Web applications & APIs
- **Fly.io** → Containerized backend
- **VPS** → Full server control